where $A, B, D > 0$ with $B \approx 4A$ (diagonal bias) and η is the learning rate. After the second step, $v_2^{(2)}$ becomes dominant, i.e. $v_2^{(2)} > v_j^{(2)}, j = 1, 3, 4, \ldots, t$.

Proof. **First step**. We analyze the first step of stochastic gradient descent. The function at initialization is

$$\begin{aligned} f^0(e)_{p,i} &= c^2 \sum_{t'=1}^{p} \mathbb{1}\{x_{t'} = i\} \sum_{s=1}^{t'} e_{x_p}^\top 1 1^T e_{x_s} \\ &= c^2 \sum_{t'=1}^{p} \mathbb{1}\{x_{t'} = i\} t' \quad \in [0, c^2 \frac{p(p+1)}{2}]. \end{aligned} \tag{A.14}$$

By choosing $\Delta \geq c^2 \frac{t(t+1)}{2}$, we ensure that $\Delta + f^0(e)_{p,i} - f^0(e)_{p,x_{p+1}} \geq 0$, for all $p \in [t]$. From the total law of expectation it is:

$$\mathop{\mathbb{E}}_{\substack{\boldsymbol{x} \sim \mathcal{P} \\ \mathcal{P} \sim \mathrm{Dir}(\boldsymbol{\alpha})^{\otimes k}}} [L] = \mathbb{E}_{\mathcal{P} \sim \mathrm{Dir}(\boldsymbol{\alpha})^{\otimes k}} \left[\mathbb{E}_{\boldsymbol{x} \sim \mathcal{P}} [L | \mathcal{P}] \right]. \tag{A.15}$$

We will first focus on the inner conditional expectation and at the end of the calculations we will take expectation with respect the transition matrix. In what follows, unless otherwise stated, $\mathbb{E}$ and $\mathbb{P}$ will be with respect to the randomness of $\boldsymbol{x}$ conditioned on $\mathcal{P}$.

Then the gradient of the loss with respect to W_k is:

$$\begin{aligned} \nabla_{W_k} \mathbb{E}_{\boldsymbol{x} \sim \mathcal{P}} [L | \mathcal{P}] &= \frac{1}{t} \sum_{p=1}^{t} \frac{1}{k} \sum_{i=1, i \neq x_{p+1}}^{k} \mathbb{E}\left[\mathbb{1}\left\{ \Delta + f(e)_{p,i} - f(e)_{p,x_{p+1}} \geq 0 \right\} \left(\nabla_{W_k} f(e)_{p,i} - \nabla_{W_k} f(e)_{p,x_{p+1}} \right) \right] \\ &= \frac{1}{t} \sum_{p=1}^{t} \frac{1}{k} \sum_{i=1, i \neq x_{p+1}}^{k} \mathbb{E}\left[\left(\nabla_{W_k} f(e)_{p,i} - \nabla_{W_k} f(e)_{p,x_{p+1}} \right) \right] \\ &= \frac{1}{t} \sum_{p=1}^{t} \frac{1}{k} \sum_{i=1}^{k} \mathbb{E}\left[\left(\nabla_{W_k} f(e)_{p,i} - \nabla_{W_k} f(e)_{p,x_{p+1}} \right) \right] \\ &= \frac{1}{t} \sum_{p=1}^{t} \left[\left(\frac{1}{k} \sum_{i=1}^{k} \mathbb{E}\left[\nabla_{W_k} f(e)_{p,i} \right] \right) - \mathbb{E}\left[\nabla_{W_k} f(e)_{p,x_{p+1}} \right] \right]. \end{aligned} \tag{A.16}$$

From equation (A.10), we have for the gradient of logit i:

$$\nabla_{W_k} f(e)_{p,i} = c \sum_{t'=1}^{p} \mathbb{1}\{x_{t'} = i\} \sum_{s=1}^{t'} e_{x_p} e_{x_s}^T, \tag{A.17}$$

or, equivalently, its elements are:

$$\left(\nabla_{W_k} f(e)_{p,i} \right)_{m,l} = c \sum_{t'=1}^{p} \mathbb{1}\{x_{t'} = i\} \sum_{s=1}^{t'} \mathbb{1}\{x_p = m\} \mathbb{1}\{x_s = l\}, \tag{A.18}$$

and their expectation (with respect to $\boldsymbol{x}$ conditioned on $\mathcal{P}$) is:

$$\mathbb{E}\left[\left(\nabla_{W_k} f(e)_{p,i} \right)_{m,l} \right] = c \sum_{t'=1}^{p} \sum_{s=1}^{t'} \mathbb{P}\left[x_s = l, x_{t'} = i, x_p = m \right]. \tag{A.19}$$

Similarly, for the ground truth logit:

$$\mathbb{E}\left[\left(\nabla_{W_k} f(e)_{p,x_{p+1}} \right)_{m,l} \right] = c \sum_{t'=1}^{p} \sum_{s=1}^{t'} \mathbb{P}\left[x_s = l, x_{t'} = x_{p+1}, x_p = m \right]. \tag{A.20}$$

Thus, by substituting back to eq. (A.16) we have :

$$
\begin{aligned}
-\nabla_{W_k}\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}}\left[L|\mathcal{P}\right] &= \frac{c}{t}\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\mathbb{P}\left[x_s=l, x_{t'}=x_{p+1}, x_p=m\right] - \frac{c}{tk}\sum_{i=1}^{k}\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\mathbb{P}\left[x_s=l, x_{t'}=i, x_p=m\right] \\
&= \frac{c}{t}\sum_{i=1}^{k}\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\mathbb{P}\left[x_{p+1}=i, x_s=l, x_{t'}=i, x_p=m\right] - \frac{c}{tk}\sum_{i=1}^{k}\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\mathbb{P}\left[x_s=l, x_{t'}=i, x_p=m\right] \\
&= \frac{c}{t}\sum_{i=1}^{k}\left(\mathcal{P}_{mi}-\frac{1}{k}\right)\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\mathbb{P}\left[x_s=l, x_{t'}=i, x_p=m\right] \\
&= \frac{c}{t}\sum_{i=1}^{k}\left(\mathcal{P}_{mi}-\frac{1}{k}\right)\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\mathbb{P}\left[x_p=m \mid x_{t'}=i\right]\mathbb{P}\left[x_{t'}=i \mid x_s=l\right]\mathbb{P}\left[x_s=l\right] \\
&= \frac{c}{t}\sum_{i=1}^{k}\left(\mathcal{P}_{mi}-\frac{1}{k}\right)\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\left(\mathcal{P}^{p-t'}\right)_{im}\left(\mathcal{P}^{t'-s}\right)_{li}\pi_l,
\end{aligned}
\tag{A.21}
$$

where we used the Markov property and the assumption that the chain has reached its stationary distribution $\boldsymbol{\pi}\in\mathbb{R}^{1\times k}$, so $\mathbb{P}\left[x_t=i\right]=\pi_i$ for all $t>0, i\in[k]$. Recall that, in the case of $k=2$, the (random) transition matrix can be parameterized as:

$$
\mathcal{P}=\begin{pmatrix} a & 1-a \\ 1-b & b \end{pmatrix}, \tag{A.22}
$$

where $a,b\sim Unif([0,1])$. $\mathcal{P}$ is almost surely diagonalizable, and using its eigendecomposition, we can calculate its n-th powers

$$
\begin{aligned}
\mathcal{P}^n &= \frac{1}{a+b-2}\begin{pmatrix} 1 & \frac{1-a}{b-1} \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & a+b-1 \end{pmatrix}^n\begin{pmatrix} b-1 & a-1 \\ 1-b & b-1 \end{pmatrix} \\
&= \frac{1}{a+b-2}\begin{pmatrix} (a-1)(a+b-1)^n+b-1 & (1-a)(a+b-1)^n+a-1 \\ (1-b)(a+b-1)^n+b-1 & (b-1)(a+b-1)^n+a-1 \end{pmatrix}.
\end{aligned}
\tag{A.23}
$$

We observe that every element of the matrix, $\left(\mathcal{P}^n\right)_{ij}$, is of the form $\frac{\beta_{ij}\lambda^n+\gamma_{ij}}{\lambda-1}$, with $\lambda=a+b-1$. It is also $\boldsymbol{\pi}=\frac{1}{\lambda-1}\begin{pmatrix} b-1 & a-1 \end{pmatrix}$ (the normalized eigenvector that corresponds to λ). Thus, eq. (A.21) can be written as

$$
\begin{aligned}
-\nabla_{W_k}\mathbb{E}_{\boldsymbol{x}\sim\mathcal{P}}\left[L|\mathcal{P}\right] &= \frac{c}{t(\lambda-1)^2}\sum_{i=1}^{2}\left(\mathcal{P}_{mi}-\frac{1}{2}\right)\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\left(\beta_{im}\lambda^{p-t'}+\gamma_{im}\right)\left(\beta_{li}\lambda^{t'-s}+\gamma_{li}\right)\pi_l \\
&= \frac{c\pi_l}{t(\lambda-1)^2}\sum_{i=1}^{2}\left(\mathcal{P}_{mi}-\frac{1}{2}\right)\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'}\left(\underbrace{\beta_{im}\beta_{li}\lambda^{p-s}}_{(A)}+\underbrace{\beta_{im}\gamma_{li}\lambda^{p-t'}}_{(B)}+\underbrace{\beta_{li}\gamma_{im}\lambda^{t'-s}}_{(C)}+\underbrace{\gamma_{im}\gamma_{li}}_{(D)}\right).
\end{aligned}
\tag{A.24}
$$

We calculate the four terms separately:

$$
\begin{aligned}
(A) &= \sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'} \beta_{im}\beta_{li}\lambda^{p-s} \\
&= \beta_{im}\beta_{li}\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'} \lambda^{p-s} \\
&= \beta_{im}\beta_{li}\sum_{p=1}^{t}\sum_{t'=1}^{p} \frac{\lambda^{p-1}-\lambda^{p-t'-1}}{1-\lambda^{-1}} \\
&= \frac{\beta_{im}\beta_{li}}{1-\lambda^{-1}}\sum_{p=1}^{t}\left(p\lambda^{p-1}-\frac{\lambda^{p-2}-\lambda^{-2}}{1-\lambda^{-1}}\right) \\
&= \frac{\beta_{im}\beta_{li}}{1-\lambda^{-1}}\left(\frac{1-(t+1)\lambda^{t}+t\lambda^{t+1}}{(1-\lambda)^2}-\frac{\lambda^{-1}-\lambda^{t-1}}{(1-\lambda^{-1})(1-\lambda)}+t\frac{\lambda^{-2}}{1-\lambda^{-1}}\right) \\
&= \frac{\beta_{im}\beta_{li}}{\lambda-1}\left(\frac{\lambda-(t+1)\lambda^{t+1}+t\lambda^{t+2}}{(1-\lambda)^2}-\frac{1-\lambda^{t}}{(1-\lambda^{-1})(1-\lambda)}+t\frac{1}{\lambda-1}\right).
\end{aligned}
$$

$$
\begin{aligned}
(B) &= \sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'} \beta_{im}\gamma_{li}\lambda^{p-t'} \\
&= \beta_{im}\gamma_{li}\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'} \lambda^{p-t'} \\
&= \beta_{im}\gamma_{li}\sum_{p=1}^{t}\sum_{t'=1}^{p} t'\lambda^{p-t'} \\
&= \beta_{im}\gamma_{li}\sum_{p=1}^{t}\lambda^{p-1}\frac{1-(p+1)\lambda^{-p}+p\lambda^{-p-1}}{(1-\lambda^{-1})^2} \\
&= \beta_{im}\gamma_{li}\sum_{p=1}^{t}\frac{\lambda^{p-1}-(p+1)\lambda^{-1}+p\lambda^{-2}}{(1-\lambda^{-1})^2} \\
&= \frac{\beta_{im}\gamma_{li}}{(1-\lambda^{-1})^2}\left(\frac{1-\lambda^{t}}{1-\lambda}-\lambda^{-1}\frac{t(t+3)}{2}+\lambda^{-2}\frac{t(t+1)}{2}\right) \\
&= \frac{\beta_{im}\gamma_{li}}{(\lambda-1)^2}\left(\lambda^{2}\frac{1-\lambda^{t}}{1-\lambda}-\lambda\frac{t(t+3)}{2}+\frac{t(t+1)}{2}\right).
\end{aligned}
$$

$$
\begin{aligned}
(C) &= \sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'} \beta_{li}\gamma_{im}\lambda^{t'-s} \\
&= \beta_{li}\gamma_{im}\sum_{p=1}^{t}\sum_{t'=1}^{p}\sum_{s=1}^{t'} \lambda^{t'-s} \\
&= \beta_{li}\gamma_{im}\sum_{p=1}^{t}\sum_{t'=1}^{p} \frac{\lambda^{t'-1}-\lambda^{-1}}{1-\lambda^{-1}} \\
&= \frac{\beta_{li}\gamma_{im}}{1-\lambda^{-1}}\sum_{p=1}^{t}\left(\frac{1-\lambda^{p}}{1-\lambda}-\lambda^{-1}p\right) \\
&= \frac{\beta_{li}\gamma_{im}}{1-\lambda^{-1}}\left(\frac{t}{1-\lambda}-\frac{\lambda-\lambda^{t+1}}{(1-\lambda)^2}-\lambda^{-1}\frac{t(t+1)}{2}\right) \\
&= \frac{\beta_{li}\gamma_{im}}{\lambda-1}\left(t\frac{\lambda}{1-\lambda}-\frac{\lambda^{2}-\lambda^{t+2}}{(1-\lambda)^2}-\frac{t(t+1)}{2}\right)
\end{aligned}
$$